

RESEARCH ARTICLE/REVIEW

A Data-Driven Study on Bone Deficiency and Osteoporosis Risk in Women

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Abstract: Abstract: Bone deficiency is becoming a major health issue for women. It is closely linked to hormonal changes, poor nutrition, and lifestyle factors. Events like pregnancy, lactation, and menopause speed up the loss of bone mineral density. Inadequate intake of calcium and vitamin D further harms bone health. Since early bone deficiency usually does not show visible symptoms, it often goes undiagnosed until serious problems arise. This study aims to find early warning signs of bone deficiency among women using a data-driven approach. By looking at health, nutrition, and lifestyle factors, the study seeks to promote early detection and preventive measures. The results can help healthcare planning by lowering osteoporosis risk, improving screening methods, and supporting better long-term bone health for women.

Keywords: Bone Deficiency, Women's Health, Osteoporosis, Bone Mineral Density, Hormonal Changes, Nutritional Deficiency, Early Detection

1. Introduction

Bone health plays a vital role in maintaining overall physical strength and quality of life, particularly among women who are more vulnerable to bone-related disorders [1]. Across different stages of life, women experience biological changes that significantly affect bone metabolism. Periods such as pregnancy, lactation, and menopause introduce hormonal fluctuations that accelerate bone mineral loss [2]. When these physiological changes are combined with poor nutrition, inadequate calcium and vitamin D intake, and limited physical activity, the risk of bone deficiency increases gradually over time [3].

Despite its seriousness, bone deficiency often progresses silently and remains unnoticed in its early stages [4]. Many women are unaware of declining bone health until fractures, chronic pain, or mobility limitations appear. Common health conditions such as anemia, hormonal imbalance, and thyroid disorders further contribute to weakened bone structure, yet their impact on bone health is frequently overlooked [5]. Limited awareness, irregular health check-ups, and restricted access to healthcare services delay timely diagnosis and intervention [6].

Osteoporosis and related bone disorders impose long-term physical, psychological, and economic burdens on women and healthcare systems [7]. Early identification of risk factors is therefore essential to prevent severe complications. A data-driven approach that integrates health, nutritional, and lifestyle information can provide valuable insights for early detection and preventive care [8]. This study aims to contribute to this approach by systematically analyzing factors associated with bone deficiency among women.

2. Problem Statement

Women are at a higher risk of bone deficiency due to hormonal changes, nutritional deficiencies, and lifestyle-related factors. Life stages such as pregnancy, lactation, and menopause significantly accelerate the reduction of bone mineral density. Inadequate intake of essential nutrients, particularly calcium and vitamin D, along with low physical activity, further weakens bone health. Early stages of bone deficiency often remain undetected because symptoms are not immediately visible. Common health issues among women, including anemia and hormonal imbalance, indirectly affect bone strength and delay diagnosis. The absence of regular screening and data-driven risk identification leads to late intervention, increasing the likelihood of osteoporosis, fractures, and long-term health complications.

2.1 Objective Of The Study

1. To examine variations in bone mineral density among women across different life stages such as adolescence, pregnancy, lactation, and menopause. Each stage represents distinct biological and hormonal changes that influence bone strength over time and do not necessarily follow a uniform progression.
2. To analyze the influence of nutritional factors, particularly calcium and vitamin D intake, on bone health in women. Bone mineral density responds gradually to long-term dietary habits, and nutritional deficiencies often develop silently before clinical symptoms appear.
3. To identify the relationship between commonly observed health conditions in women, such as anemia, hormonal imbalance, and thyroid disorders, and their impact on bone mineral density and overall skeletal health.
4. To study the role of lifestyle-related factors, including physical activity levels, daily routines, and sedentary behavior, in the maintenance or deterioration of bone health among women over time.
5. To observe patterns of bone deficiency using available health records and screening data in order to recognize early risk indicators before severe complications such as osteoporosis and fractures develop. Data-based observation supports timely and informed decision making.
6. To support preventive healthcare strategies by generating data-driven insights that can assist in early screening, timely intervention, and long-term improvement of bone health outcomes among women.

2.2 Hypothesis

H0 (Null Hypothesis)

Bone deficiency in women is not significantly influenced by hormonal changes, nutritional intake, or lifestyle-related factors. Variations in bone mineral density occur independently of calcium and vitamin D intake, physical activity levels, or health conditions such as anemia and hormonal imbalance.

H1 (Alternative Hypothesis)

Bone deficiency in women is significantly influenced by hormonal changes, nutritional intake, and lifestyle-related factors. Variations in bone mineral density are closely associated with calcium and vitamin D intake, physical activity patterns, and related health conditions.

H0₂ (Null Hypothesis)

Nutritional deficiencies, including low calcium and vitamin D intake, do not have a measurable effect on bone mineral density in women. Bone strength remains largely unchanged despite variations in nutritional levels.

H1₂ (Alternative Hypothesis)

Nutritional deficiencies, particularly inadequate intake of calcium and vitamin D, play a significant role in reducing bone mineral density in women. Adequate and balanced nutrient intake is essential for maintaining healthy bone structure.

H0₃ (Null Hypothesis)

Lifestyle factors such as physical inactivity and irregular health checkups do not significantly affect bone health outcomes in women. Bone deficiency occurs regardless of daily activity patterns or preventive health practices.

H1₃ (Alternative Hypothesis)

Lifestyle factors such as physical inactivity and lack of regular health screening significantly contribute to bone deficiency in women. Changes in daily activity levels and preventive behavior directly influence bone health outcomes.

3. Literature Review

Bone deficiency and osteoporosis have become major public health concerns, particularly among women, due to their strong association with hormonal, nutritional, and lifestyle-related factors. Existing literature consistently reports that women experience a higher risk of reduced bone mineral density compared to men, especially during critical life stages such as pregnancy, lactation, and menopause [1]. Hormonal changes during these periods accelerate bone resorption, making women more vulnerable to long-term skeletal weakening.

Several studies highlight that bone health in women is dynamic and varies across the lifespan. Research focusing on postmenopausal women identifies estrogen deficiency as a primary contributor to rapid bone loss, often leading to osteopenia and osteoporosis if preventive measures are not taken [2], [3]. Population-based studies conducted across different regions reveal a high prevalence of low bone mineral density among middle-aged and elderly women, suggesting that bone loss often begins well before clinical symptoms become evident [4]. This silent progression underscores the importance of early identification of risk factors.

Nutritional intake has been widely recognized as a critical determinant of bone health in women. Numerous studies report a strong association between inadequate calcium intake and reduced bone mineral density [5]. Vitamin D deficiency further aggravates this condition by limiting calcium absorption, thereby accelerating bone loss [6]. Longitudinal research indicates that nutritional deficiencies tend to develop gradually and remain unnoticed until significant deterioration is detected through bone density screening [7]. These findings emphasize the need for sustained dietary monitoring rather than short-term nutritional interventions.

In addition to nutrition, common health conditions prevalent among women have been shown to influence bone metabolism. Studies report that anemia is associated with lower bone mineral density due to reduced oxygen transport and impaired nutrient delivery at the cellular level [8], [9]. Hormonal imbalances and thyroid disorders have also been linked to altered bone remodeling processes, increasing fracture risk when left unmanaged [10]. Such findings highlight the interconnected nature of bone health with broader physiological conditions affecting women.

Lifestyle-related factors play a significant role in either preserving or weakening bone health. Physical inactivity and sedentary behavior are frequently cited as contributors to bone loss, particularly among urban and aging female populations [11]. Comparative studies demonstrate that women engaging in regular weight-bearing and resistance exercises show higher bone mineral density compared to inactive individuals [12]. Conversely, prolonged inactivity has been shown to accelerate bone loss, especially in postmenopausal women, where mechanical stimulation is essential for maintaining skeletal integrity [13].

Preventive healthcare practices, including regular screening and early diagnosis, are also emphasized in the literature. Multiple studies indicate that lack of routine bone health screening leads to delayed diagnosis, allowing bone deficiency to progress into severe osteoporosis and fractures [14]. Women with limited access to healthcare services or low awareness of bone-related disorders are particularly vulnerable to late-stage complications [15]. These findings support the need for systematic screening programs and improved awareness initiatives.

Recent research increasingly focuses on data-driven approaches for early detection of bone deficiency. Studies utilizing bone mineral density measurements, biochemical markers, and health records demonstrate that analytical methods can effectively identify patterns associated with early bone loss [16]. Data-based risk assessment models have been shown to support timely intervention by identifying high-risk individuals before irreversible skeletal damage occurs [17]. Such approaches offer a shift from reactive treatment toward preventive healthcare strategies.

Interventional studies further support the effectiveness of combined nutritional and lifestyle modifications. Research examining calcium and vitamin D supplementation alongside regular physical activity reports measurable improvements in bone mineral density among women over time [18]. These findings suggest that bone deficiency can be managed and potentially prevented when evidence-based strategies are applied consistently.

Despite extensive research, several gaps remain in the existing literature. Many studies focus on individual factors such as nutrition or physical activity in isolation, without considering their combined influence on bone health [19]. Additionally, a large proportion of studies rely on cross-sectional designs, limiting the understanding of long-term changes in bone mineral density [20]. There is also limited integration of health conditions, lifestyle behaviors, and screening data within unified analytical frameworks, which restricts early risk prediction [21].

Global and regional analyses project a growing burden of osteoporosis among women, with increasing fracture rates and healthcare costs if preventive measures are not strengthened [22]. These studies emphasize the need to transition from late-stage treatment to early, data-driven prevention strategies. Integrating health, nutritional, and lifestyle data through analytical approaches can enhance decision-making and support effective bone health management [23].

Overall, the reviewed literature confirms that bone deficiency in women is a multifactorial condition influenced by biological, nutritional, lifestyle, and healthcare-related factors. The silent progression of bone loss, combined with limited awareness and delayed screening, contributes significantly to the high prevalence of osteoporosis among women [24]. There remains a clear need for comprehensive, data-driven studies that integrate multiple risk factors to support early detection and long-term improvement of bone health outcomes [25].

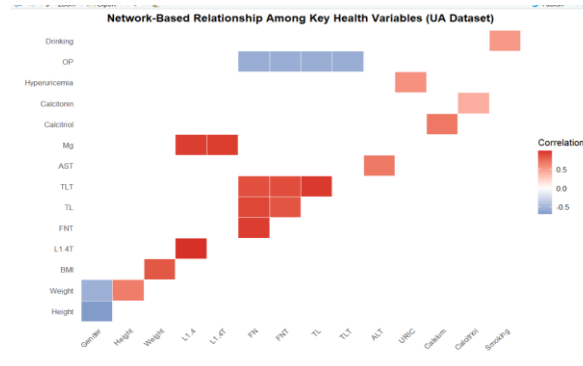


Figure 1. Network teaching quality evaluation system based on big data

3.1. Theoretical framework

Bone health behavior among women is influenced by a combination of biological, psychological, and lifestyle factors. Understanding how women perceive bone health risks and how these perceptions shape preventive behavior is important in health psychology research [11]. Several behavioral theories have been developed to explain health-related decision making, particularly those associated with disease prevention and long-term wellbeing.

The Theory of Reasoned Action (TRA) and its extension, the Theory of Planned Behavior (TPB), proposed by Ajzen, are widely used frameworks in health behavior research [12]. The TPB is based on the premise that an individual's behavior is guided by behavioral intention, which is influenced by personal attitudes, social influence, and perceived control over the behavior. This model has been effectively applied to study preventive health behaviors such as nutrition, physical activity, and participation in health screening programs.

In the context of women's bone health, the TPB helps explain how attitudes toward bone deficiency, osteoporosis prevention, and screening influence behavioral intentions. Positive attitudes toward adequate calcium intake, regular physical activity, and routine health checkups increase the likelihood of engaging in bone-protective behaviors [2]. Social expectations from family members and healthcare professionals also play an important role in shaping these behaviors.

According to the TPB, three key determinants influence intention: attitude toward the behavior, subjective norm, and perceived behavioral control [11]. Attitude reflects beliefs about the benefits of preventive actions, subjective norm represents perceived social pressure, and perceived behavioral control relates to confidence in performing the behavior despite barriers. The influence of these factors varies across individuals and contexts.

This theoretical framework supports the present study by explaining how health awareness, lifestyle choices, and preventive practices interact to influence bone health outcomes among women. It also provides a foundation for developing data-driven interventions aimed at early detection and long-term improvement of bone health.

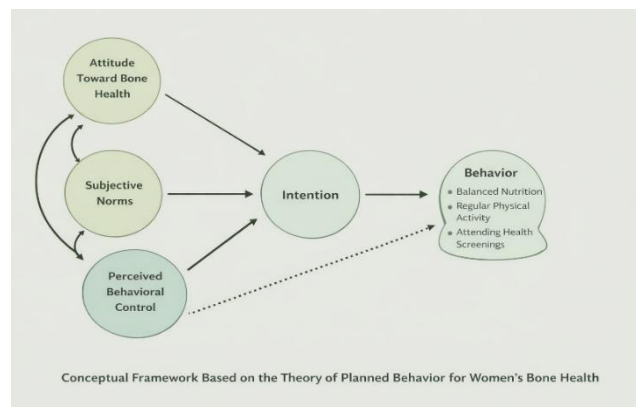


Figure 2. Theory of Planned Behaviour for Women's Bone Health Prevention

4. Research Methodology

4.1. Research design

Focusing on a single point in time, this study uses real-world health and lifestyle data to understand factors associated with bone deficiency among women. Instead of observing changes across several years, the research concentrates on current bone health status, nutritional intake, lifestyle behavior, and preventive practices as they exist together at present. By examining these variables simultaneously, the study aims to identify patterns related to bone mineral density and early risk indicators. This snapshot-based approach is commonly used in health and behavioral research as it allows efficient data collection without high cost or long-term follow-up.

A cross-sectional design provides valuable insight into relationships between multiple variables when measured at the same time, as noted by Salkind (2010) and Sedgwick (2014). Rather than tracking long-term progression, this study gathers health records, lifestyle information, and screening-related data within a single timeframe. Because all information is aligned to the same period, it becomes easier to observe how attitudes, awareness, and perceived control influence preventive bone health behavior. Guided by the Theory of Planned Behavior, this design helps explain how present beliefs and intentions shape health-related decisions among women.

5. Data Description

Bone health patterns among women were analyzed using two structured datasets obtained from open-access sources, namely *osteoporosis.csv* and *UA.csv*. These datasets were combined to examine how biological, nutritional, and lifestyle-related factors influence bone mineral density and bone health outcomes among women. Each dataset contained detailed records related to individual health characteristics, enabling a comprehensive assessment of bone deficiency risk using real-world data.

The *osteoporosis.csv* dataset primarily included information related to bone mineral density, osteoporosis status, age, physical activity, and nutritional indicators. In parallel, the *UA.csv* dataset contributed additional health-related variables, including biochemical and clinical measurements relevant to women's health. Together, these datasets allowed the examination of multiple dimensions of bone health, including skeletal strength, nutritional intake, lifestyle behavior, and associated medical conditions.

Both datasets were organized in CSV format and analyzed using statistical and graphical techniques through R software. Their structured tabular layout enabled efficient visualization and identification of trends related to bone deficiency. By integrating these datasets, patterns emerged that linked nutritional intake, physical activity, and health conditions with variations in bone mineral density. All analysis was conducted using observed data, ensuring that interpretations were based on measurable evidence rather than assumptions.

5.1 Type of Data

Quantitative Health and Screening Data

5.2 Data Source

The data used in this study was obtained from open-access datasets available on public data platforms. The *osteoporosis.csv* dataset provided detailed records related to bone mineral density and osteoporosis classification, while the *UA.csv* dataset supplied supplementary health and clinical variables associated with women's health status. Both datasets were publicly accessible and ethically suitable for academic research. All records were stored in CSV format, allowing seamless processing, statistical analysis, and visualization using R software once the data was cleaned and structured.

5.3 Key Characteristics of Dataset

1. The datasets consist of numerical values representing bone mineral density, nutritional intake, physical activity levels, and clinical health indicators collected from real-world observations.
2. Data entries were collected across multiple records, allowing the identification of patterns related to bone health and osteoporosis risk among women.
3. Bone mineral density serves as the primary outcome variable, reflecting skeletal strength and overall bone health status.

4. Nutritional, lifestyle, and health-related variables show observable relationships with bone mineral density, enabling the study of contributing risk factors.
5. The datasets are arranged in rows and columns, making them suitable for statistical analysis and graphical representation. Trends become more visible when the data is explored beyond surface-level values.
6. Pattern analysis supports the identification of bone deficiency risk, while correlation analysis helps reveal associations between contributing variables and bone health outcomes.
7. Stored in CSV format, the datasets are compatible with multiple analytical tools and environments, allowing flexible processing and reproducibility.

5.4 Data Attributes

The datasets include attributes that describe bone health and associated factors among women. Bone mineral density values represent skeletal strength and serve as the primary indicator of bone health. Nutritional attributes include daily calcium intake and vitamin D levels, which influence bone formation and calcium absorption.

Lifestyle-related attributes such as physical activity levels reflect behavioral patterns that contribute to bone maintenance or deterioration. Health-related variables including anemia status and hormonal or thyroid conditions are included due to their influence on bone metabolism and remodelling processes. Bone health outcomes are classified based on bone mineral density values, enabling clear categorization of bone deficiency and osteoporosis risk.

5.5 Dataset CSV Files

The datasets used in this study are stored in CSV format and include *UA.csv* and *osteoporosis.csv*.

6. Graph Using R language

6.1 Dataset Summery

Variable	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
Gender	1.00	1.00	1.00	1.00	1.00	1.00	3
Age	28.00	28.00	50.00	60.00	57.30	77.00	3
Height	140.0	140.00	156.80	161.00	160.18	178.00	3
Weight	42.00	42.00	57.30	61.75	60.79	75.00	3
BMI	19.213	19.213	22.286	26.240	23.562	29.805	3
L1	0.763	0.763	1.100	1.196	1.136	1.600	36
L1-4T	0.500	0.500	0.830	1.056	1.136	1.480	36
FN	0.3275	0.3275	0.6460	0.7155	0.6501	0.8690	NA's 36
FN.T	0.3800	0.3800	0.7380	0.7680	0.7127	0.7769	NA's 36
TL.T	0.7000	0.7000	1.0000	1.1000	1.1022	1.2000	3
ALT	4.00	4.00	16.25	18.50	17.90	26.00	
AST	11.00	11.00	21.75	22.00	22.60	31.00	
BUN	7.40	7.40	12.83	14.60	14.56	21.00	
CREA	0.860	0.860	0.970	1.065	1.0970	1.500	NA's 3
URIC	2.66	2.66	3.98	5.16	7.68	13.74	3
FBG	3.130	3.9000	3.0000	3.000	0.09550	0.0977	NA's 3
Calcium	7.9000	7.9000	9.0000	1.000	0.00000	1.0000	NA's 3
Calcitriol	0.0000	0.0000	0.0000	0.0500	0.560	1.0000	NA's 3
HTN	0.0000	0.0000	0.0000	0.0000	0.0400	1.0000	NA's 3

Figure 3: Summery Table Of Dataset UA.csv

6.2 Graph

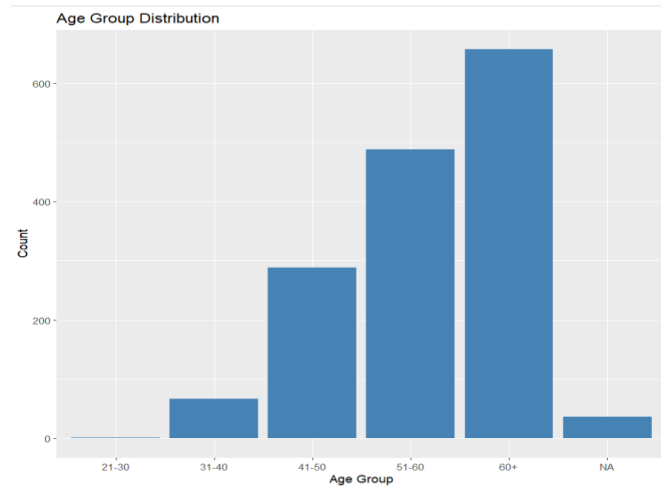


Figure.4: Age Distribution Bar Graph

7. Methodology

This study adopts a data-driven analytical approach to examine factors associated with bone deficiency among women using structured health and screening datasets. The analysis is based on quantitative health data collected over recent years, focusing on bone mineral density, nutritional intake, lifestyle behavior, and associated health conditions. R programming was used to perform descriptive statistical analysis, correlation analysis, and graphical visualization in order to understand patterns and relationships influencing women's bone health.

7.1 Data Preprocessing

Prior to analysis, the datasets were carefully reviewed to ensure accuracy and consistency. Each record was examined to identify and remove duplicate entries. Missing values were handled using appropriate estimation techniques to avoid distortion of results. After cleaning, all variables were formatted and standardized to maintain uniformity across the dataset. Outliers were reviewed and assessed for potential influence on analytical outcomes before inclusion.

Incomplete or inconsistent health records were excluded from the analysis to maintain data quality. Nutritional and lifestyle variables were standardized to comparable scales where required. Bone mineral density values and related health indicators were transformed into consistent ranges while preserving their original meaning. Relevant features related to bone health were selected for analysis, resulting in a cleaner and more reliable dataset.

7.2 Analytical Approach

A data-driven analytical approach was employed to evaluate bone health status and contributing risk factors among women. The dataset was analyzed using descriptive statistics to summarize health and lifestyle attributes. Correlation analysis was applied to identify relationships between bone mineral density and factors such as nutritional intake, physical activity, and health conditions. Graphical analysis using R was performed to visualize trends and patterns in bone health outcomes across different profiles.

7.3 Prediction Perspective

Bone health outcomes were analyzed across different health and lifestyle categories to identify patterns related to bone deficiency risk. This approach helped in classifying bone health status based on bone mineral density values and associated factors. The analysis supported early identification of individuals at higher risk of bone deficiency, enabling insights into preventive and screening-oriented strategies.

7.4 Analytical Techniques Used

Based on suitability and practical implementation, the following analytical techniques were used in this study:

- Descriptive Statistics
- Correlation Analysis
- Linear Regression
- Data Visualization using ggplot2

These techniques were selected for their effectiveness in understanding relationships between health, lifestyle, and nutritional factors and bone mineral density, and for supporting data-driven insights into women's bone health.

8. Results and Discussion

The analytical results indicate that nutritional, lifestyle, and health-related factors have a significant influence on bone health outcomes among women. Bone mineral density showed noticeable variation across different profiles, with calcium and vitamin D intake displaying a positive association with skeletal strength. Graphical trends revealed that women with balanced nutritional intake and higher physical activity levels tended to exhibit better bone health indicators, while reduced activity and nutritional deficiencies were associated with lower bone mineral density.

Correlation and regression analysis demonstrated that adequate calcium and vitamin D levels were positively related to bone mineral density, whereas conditions such as anemia and hormonal imbalance showed a negative association with bone health outcomes. Physical inactivity was also found to contribute to declining bone strength. These findings confirm that bone deficiency among women is influenced by a combination of biological, nutritional, and lifestyle factors rather than a single determinant.

The results support existing health studies that emphasize the importance of early screening and continuous monitoring of bone health in women. The study highlights that data-driven health assessment systems can act as early warning tools to identify bone deficiency risk before severe complications such as osteoporosis and fractures occur. Such approaches can support preventive healthcare strategies and improve long-term bone health outcomes among women.

9. Conclusion

Bone health among women is closely linked to nutritional intake, lifestyle behavior, and underlying health conditions. Traditional approaches that rely on delayed diagnosis and visible symptoms often result in late-stage detection of bone deficiency. In contrast, data-driven analysis enables early identification of risk factors and supports timely preventive intervention. By integrating health, nutritional, and lifestyle data, this study demonstrates the value of systematic assessment in understanding bone deficiency patterns. Such an approach promotes informed decision making, reduces long-term health risks, and contributes to improved quality of life for women.

Recommendations

The findings of this study suggest that increased awareness and regular screening are essential for improving bone health outcomes among women. Nutritional education focusing on adequate calcium and vitamin D intake, along with encouragement of regular physical activity, can help reduce the risk of bone deficiency. Healthcare providers and public health initiatives should promote early screening and preventive practices to support long-term skeletal health among women.

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Ethical Statement

This study did not involve any experimental procedures on human participants or animals. All analyses were carried out using publicly available secondary datasets and previously published research, ensuring compliance with ethical research standards.

Conflicts of Interest

The author declares that there is no conflict of interest regarding the publication of this research.

Data Availability Statement

The datasets analyzed during the current study are publicly available and were obtained from open-access sources. No new data were created as part of this research.

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